

Note: This question paper contains two parts A and B.

i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART- A**(10 Marks)**

- Write the following statement in symbolic form: "the crop will be destroyed if there is a flood". [1]
- Write the following statement in symbolic form: All the first year students know C Programming. Manisha is a first year student. [1]
- Define power set. [1]
- Prove that $A - (A - B) = A \cap B$. [1]
- State modular lattice. [1]
- State monoid homomorphism. [1]
- How many ways can we draw a club and a diamond from a pack of cards? [1]
- Find the number of non-negative integral solutions to $n_1 + n_2 + n_3 + n_4 = 20$. [1]
- Does there exist an Eulerian graph with an even number of vertices and an odd number of edges. [1]
- Show that a simple connected planar graph with 17 edges and 10 vertices cannot be colored with 2-colors. [1]

PART-B**(50 Marks)**

- Construct truth table for the following $p \rightarrow (\sim q \vee r)$.
- Show that $\sim P$ is a valid inference from the premises $\sim (P \wedge \sim Q)$, $\sim Q \vee R$, $\sim R$. [5+5]

OR

- Obtain PCNF and PDNF for the following formula

$$\{p \rightarrow (q \wedge r)\} \wedge \{\sim p \rightarrow (\sim q \wedge \sim r)\}. \quad [10]$$

- Show that $(A \cap (B \cup \bar{A})) \cup B = B$
- If $A = \{1, 3, 5, 7, 8, 9\}$, $B = \{3, 5, 8\}$, then verify that $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.

OR

**Access Full Paper with Solutions on
the Filo App**

Download Filo App





The Largest Collection of Previous Year Question Papers in India



50,000+ Colleges

- ✓ **2,00,000+** Past Papers
- ✓ **3,500+** Subjects Covered
- ✓ **60,00,000+** Solved Questions
- ✓ Trusted by **2,00,000+** Students

